

GR&R; Study Report

Gauge Repeatability & Reproducibility -- AIAG MSA 4th Edition | 10 Parts | 3 Operators | 3 Trials

Equipment / Gage:	Mitutoyo 293-340-30 SN-0042
Study Performed By:	QE Team
Report Date:	2026-05-10
Study Design:	3 Operators, AIAG Crossed GR&R
Regulatory Reference:	21 CFR 820.72 AIAG MSA 4th Edition

Variation Components (%Study Variation)

Component	Study Variation	% of TV	Assessment
EV -- Equipment (Repeatability)	0.00872	10.1%	Moderate
AV -- Appraiser (Reproducibility)	0.01190	13.7%	Moderate
GR&R -- Combined	0.01475	17.0%	Marginal
PV -- Part Variation	0.08541	98.5%	--
TV -- Total Variation	0.08668	100.0%	--

Assessment column: GR&R; row uses AIAG pass/fail criteria ($\leq 10\%$ Acceptable / $10\text{-}30\%$ Marginal / $>30\%$ Unacceptable). EV and AV rows show diagnostic level (Low / Moderate / High) to identify the dominant variation source.

Number of Distinct Categories (NDC): 8 -- Adequate: gauge resolves ≥ 5 distinct categories of part variation. [PASS]

Intermediate Calculation Values

Parameter	Symbol	Value	Formula
Grand Mean	X-bar-bar	9.99454	Mean of all measurements
Avg Range (overall)	R-bar-bar	0.00286	Mean of per-operator R-bars
Upper Control Limit (R)	UCL_R	0.00736	$D4(2.574) \times R\text{-bar-bar}$
Operator Mean Spread	X-diff	0.00445	$\max(\text{op X-bar}) - \min(\text{op X-bar})$
Part Range	Rp	0.05272	$\max(\text{part avg}) - \min(\text{part avg})$
K1 constant (EV)	K1	3.05	$5.15 / d2^*(3 \text{ trials})$
K2 constant (AV)	K2	2.7	$5.15 / d2^*(3 \text{ operators})$
K3 constant (PV)	K3	1.62	$5.15 / d2^*(10 \text{ parts})$

Per-Operator Breakdown

Operator	Grand Mean (X-bar)	Avg Range (R-bar)
Alice	9.99375	0.00331
Bob	9.99715	0.00227

Operator	Grand Mean (X-bar)	Avg Range (R-bar)
Carol	9.99271	0.00300

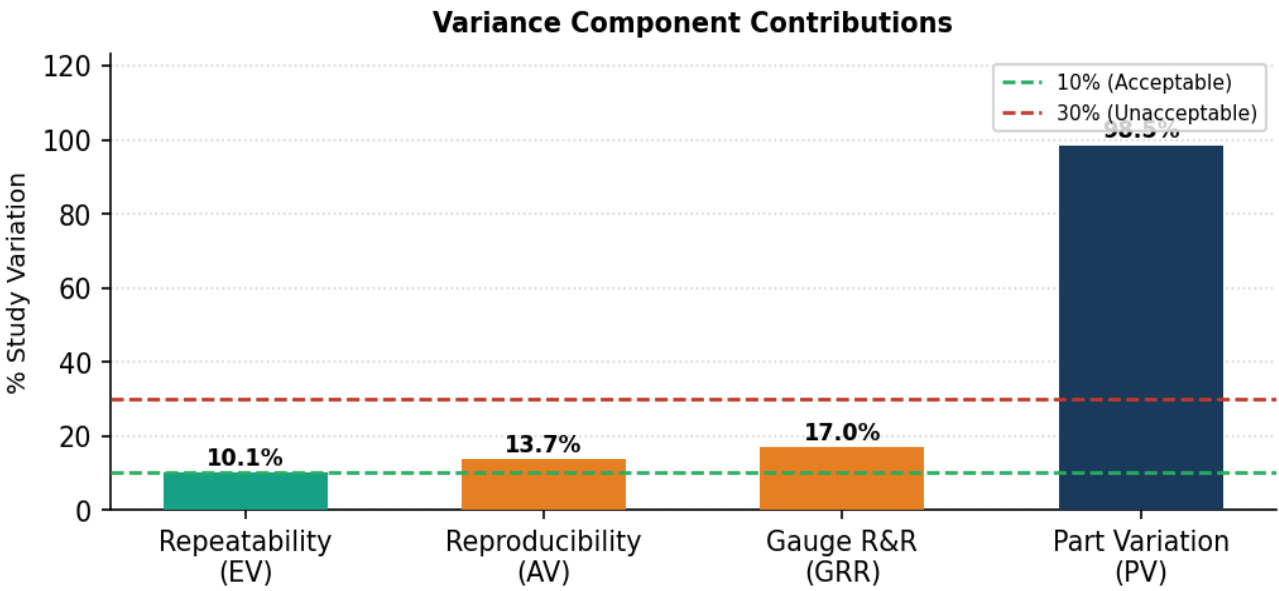
Acceptance Determination

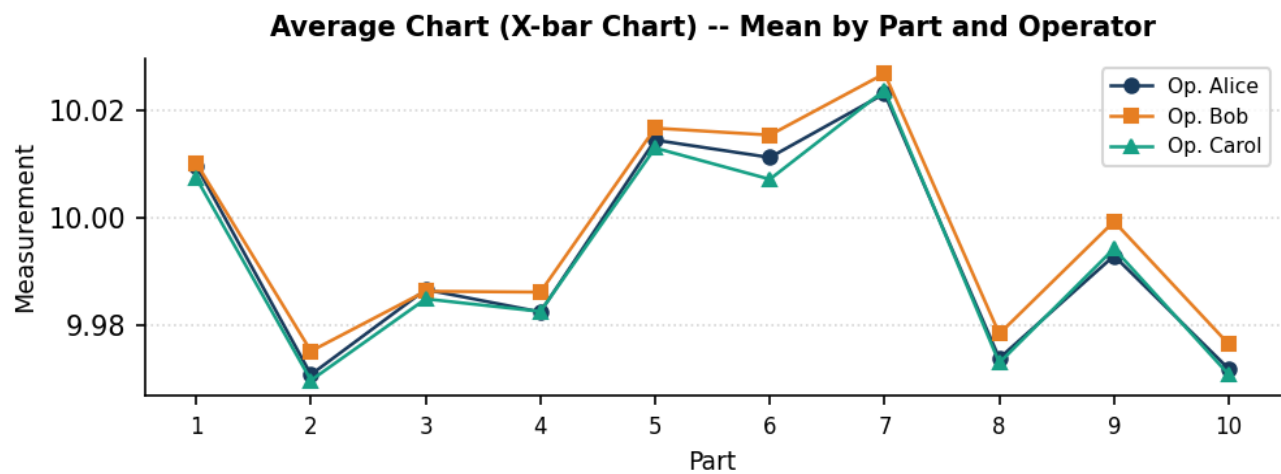
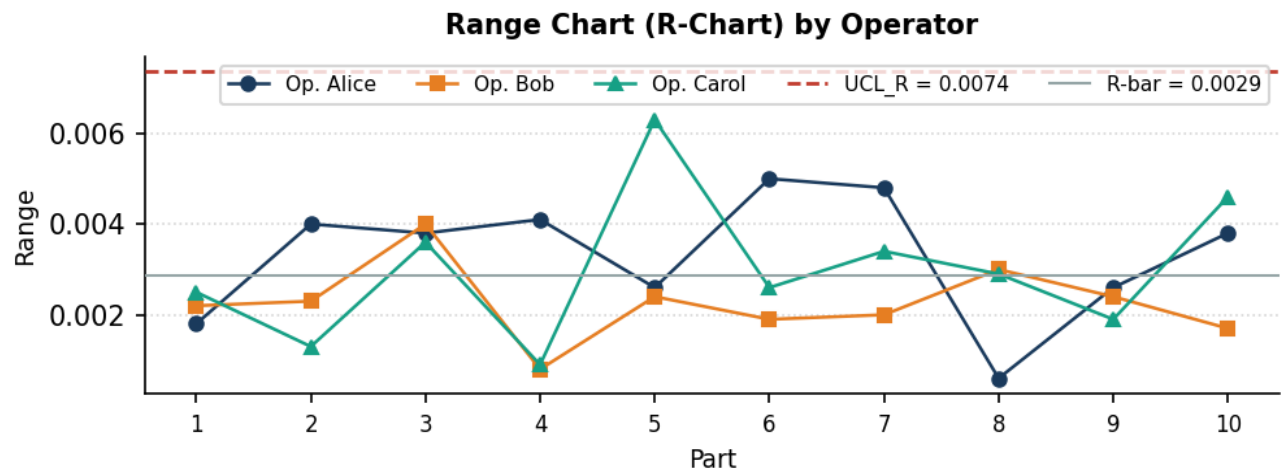
[CAUTION] MARGINAL (%GRR = 17.0% | NDC = 8)

AIAG MSA Acceptance Criteria Reference

%GRR	Assessment	Recommended Action
<= 10%	ACCEPTABLE	Gauge approved. Measurement system is capable of product acceptance decisions.
10% - 30%	MARGINAL	May be acceptable depending on importance, cost, and customer concurrence. Investigate dominant variation source.
> 30%	UNACCEPTABLE	Do not use for product acceptance. Identify root cause and correct before re-study. Check gage discrimination, comp

Control Charts





The measurement system is MARGINAL ($10\% < \%GR\&R; \leq 30\%$). Use with caution -- approval should consider application importance, cost of gage improvement, and customer concurrence. Compare EV vs. AV to identify the dominant source: if $AV > EV$, focus on operator training and fixture standardization; if $EV > AV$, evaluate gage resolution or replacement.

Generated by `grr_tool.py` -- AIAG MSA 4th Edition Average & Range Method. Study variation expressed as 5.15-sigma (99% coverage). $\%Study\ Variation = 100 \times (\text{component} / TV)$. Regulatory basis: 21 CFR Part 820.72 (FDA QSR). This report is a controlled quality record -- retain per document control procedures.